

RW4 Sheet Module - EU

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

Input workbook: full_uah.xlsx

Source sheet: Europe

Rows (clean): 111299

Date range: 2021-12-30 .. 2026-01-15

Products: 5

Stats rows: 15

Tests rows: 15

Interpretation option: use tests table -> recommended_action/recommended_model_family per product.

Table: clean

source	date	raw_product	product	standardized_type	brand	region	price	unit_ok	subcategory	unit_family	liter_equiv_allowed	maturation_type	mapping_quality_flag	matched_pattern	comparison_family	price_variant	reconstruction_variant	observed_price	baseline_price	fat_band	pack_band	brand_normalized	segment_key	shock_dummy	admissible_for_level_model	admissibility_reason
EU	2021-12-30	WMP	Інше/невідомо	other	EU	131.262525	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	131.262525	131.262525	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	EMMENTAL	Сир твердий	hard_cheese	EU	161.956644993	1	hard_cheese	mass		0	aged_hard	matched	emmental	mass_dairy	observed	observed	161.956644993	161.956644993	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	SMP	Інше/невідомо	other	EU	103.64797827	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	103.64797827	103.64797827	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	DRINKING MILK	Молоко питне	milk	EU	21.449223144	1	milk	liquid		1	fresh	multi_match	drinking milk	liquid_dairy	observed	observed	21.449223144	21.449223144	unknown	unknown	milk milk unknown unknown...	0		1	uah_per_kg
EU	2021-12-30	WMP	Інше/невідомо	other	EU	131.880231	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	131.880231	131.880231	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	WMP	Інше/невідомо	other	EU	129.83253561	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	129.83253561	129.83253561	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	EDAM	Сир твердий	hard_cheese	EU	114.75124362	1	hard_cheese	mass		0	aged_hard	matched	edam	mass_dairy	observed	observed	114.75124362	114.75124362	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	BUTTER	Масло вершкове	butter	EU	194.9601206376	1	butter	mass		0	butter	matched	butter	mass_dairy	observed	observed	194.9601206376	194.9601206376	unknown	unknown	butter butter unknown unk...	0		1	uah_per_kg
EU	2021-12-30	WHEYPOWDER	Інше/невідомо	other	EU	37.680066	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	37.680066	37.680066	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	SMP	Інше/невідомо	other	EU	106.554285	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	106.554285	106.554285	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	CREAM	Вершки	cream	EU	77.8080082221	1	cream	liquid		1	fresh	matched	cream	liquid_dairy	observed	observed	77.8080082221	77.8080082221	unknown	unknown	cream cream unknown unkno...	0		1	uah_per_kg
EU	2021-12-30	BUTTER	Масло вершкове	butter	EU	189.018036	1	butter	mass		0	butter	matched	butter	mass_dairy	observed	observed	189.018036	189.018036	unknown	unknown	butter butter unknown unk...	0		1	uah_per_kg
EU	2021-12-30	DRINKING MILK	Молоко питне	milk	EU	15.5711019627	1	milk	liquid		1	fresh	multi_match	drinking milk	liquid_dairy	observed	observed	15.5711019627	15.5711019627	unknown	unknown	milk milk unknown unknown...	0		1	uah_per_kg
EU	2021-12-30	DRINKING MILK	Молоко питне	milk	EU	13.7691300195	1	milk	liquid		1	fresh	multi_match	drinking milk	liquid_dairy	observed	observed	13.7691300195	13.7691300195	unknown	unknown	milk milk unknown unknown...	0		1	uah_per_kg
EU	2021-12-30	EMMENTAL	Сир твердий	hard_cheese	EU	139.292703	1	hard_cheese	mass		0	aged_hard	matched	emmental	mass_dairy	observed	observed	139.292703	139.292703	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	BUTTER	Масло вершкове	butter	EU	160.48310733	1	butter	mass		0	butter	matched	butter	mass_dairy	observed	observed	160.48310733	160.48310733	unknown	unknown	butter butter unknown unk...	0		1	uah_per_kg
EU	2021-12-30	EMMENTAL	Сир твердий	hard_cheese	EU	202.342263273	1	hard_cheese	mass		0	aged_hard	matched	emmental	mass_dairy	observed	observed	202.342263273	202.342263273	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	EDAM	Сир твердий	hard_cheese	EU	125.8918492977	1	hard_cheese	mass		0	aged_hard	matched	edam	mass_dairy	observed	observed	125.8918492977	125.8918492977	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	BUTTER	Масло вершкове	butter	EU	176.663916	1	butter	mass		0	butter	matched	butter	mass_dairy	observed	observed	176.663916	176.663916	unknown	unknown	butter butter unknown unk...	0		1	uah_per_kg
EU	2021-12-30	WMP	Інше/невідомо	other	EU	115.00450308	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	115.00450308	115.00450308	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	SMP	Інше/невідомо	other	EU	91.90847574	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	91.90847574	91.90847574	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	EDAM	Сир твердий	hard_cheese	EU	118.25672517	1	hard_cheese	mass		0	aged_hard	matched	edam	mass_dairy	observed	observed	118.25672517	118.25672517	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	BUTTER	Масло вершкове	butter	EU	189.39792519	1	butter	mass		0	butter	matched	butter	mass_dairy	observed	observed	189.39792519	189.39792519	unknown	unknown	butter butter unknown unk...	0		1	uah_per_kg
EU	2021-12-30	WHEYPOWDER	Інше/невідомо	other	EU	33.81631497	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	33.81631497	33.81631497	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	BUTTER	Масло вершкове	butter	EU	177.4044219528	1	butter	mass		0	butter	matched	butter	mass_dairy	observed	observed	177.4044219528	177.4044219528	unknown	unknown	butter butter unknown unk...	0		1	uah_per_kg
EU	2021-12-30	BUTTER	Масло вершкове	butter	EU	179.443593	1	butter	mass		0	butter	matched	butter	mass_dairy	observed	observed	179.443593	179.443593	unknown	unknown	butter butter unknown unk...	0		1	uah_per_kg
EU	2021-12-30	SMP	Інше/невідомо	other	EU	100.867374711	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	100.867374711	100.867374711	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	GOUDA	Сир твердий	hard_cheese	EU	126.7964797347	1	hard_cheese	mass		0	aged_hard	matched	gouda	mass_dairy	observed	observed	126.7964797347	126.7964797347	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	BUTTER	Масло вершкове	butter	EU	158.750442	1	butter	mass		0	butter	matched	butter	mass_dairy	observed	observed	158.750442	158.750442	unknown	unknown	butter butter unknown unk...	0		1	uah_per_kg
EU	2021-12-30	CHEDDAR	Сир твердий	hard_cheese	EU	103.41633852	1	hard_cheese	mass		0	aged_hard	matched	cheddar	mass_dairy	observed	observed	103.41633852	103.41633852	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	EMMENTAL	Сир твердий	hard_cheese	EU	158.2641220662	1	hard_cheese	mass		0	aged_hard	matched	emmental	mass_dairy	observed	observed	158.2641220662	158.2641220662	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	EDAM	Сир твердий	hard_cheese	EU	180.49678173	1	hard_cheese	mass		0	aged_hard	matched	edam	mass_dairy	observed	observed	180.49678173	180.49678173	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	SMP	Інше/невідомо	other	EU	96.65900550360001	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	96.65900550360001	96.65900550360001	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	CREAM	Вершки	cream	EU	82.6356585798	1	cream	liquid		1	fresh	matched	cream	liquid_dairy	observed	observed	82.6356585798	82.6356585798	unknown	unknown	cream cream unknown unkno...	0		1	uah_per_kg
EU	2021-12-30	GOUDA	Сир твердий	hard_cheese	EU	110.97397143	1	hard_cheese	mass		0	aged_hard	matched	gouda	mass_dairy	observed	observed	110.97397143	110.97397143	unknown	unknown	hard_cheese hard_cheese u...	0		1	uah_per_kg
EU	2021-12-30	WHEYPOWDER	Інше/невідомо	other	EU	30.90691971	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	30.90691971	30.90691971	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg
EU	2021-12-30	BUTTER	Масло вершкове	butter	EU	166.17835665	1	butter	mass		0	butter	matched	butter	mass_dairy	observed	observed	166.17835665	166.17835665	unknown	unknown	butter butter unknown unk...	0		1	uah_per_kg
EU	2021-12-30	CREAM	Вершки	cream	EU	87.07183776000001	1	cream	liquid		1	fresh	matched	cream	liquid_dairy	observed	observed	87.07183776000001	87.07183776000001	unknown	unknown	cream cream unknown unkno...	0		1	uah_per_kg
EU	2021-12-30	DRINKING MILK	Молоко питне	milk	EU	15.666877278	1	milk	liquid		1	fresh	multi_match	drinking milk	liquid_dairy	observed	observed	15.666877278	15.666877278	unknown	unknown	milk milk unknown unknown...	0		1	uah_per_kg
EU	2021-12-30	WMP	Інше/невідомо	other	EU	110.62187901	1	unknown	unknown		0	unknown	unmatched		other	observed	observed	110.62187901	110.62187901	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg

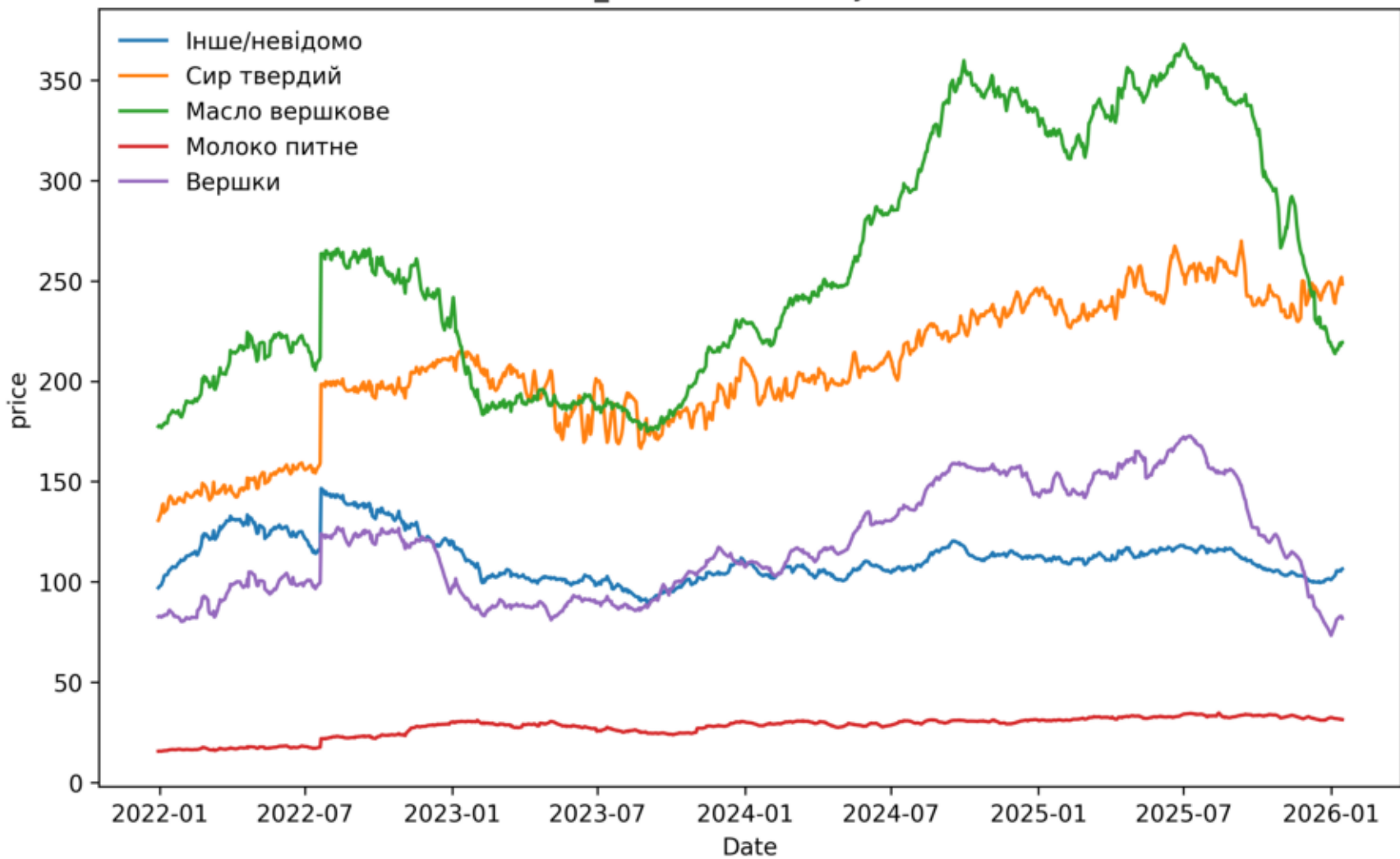
Table: descriptive_stats

source	product	standardized_type	series_variant	count	missing	imputed_share	mean	median	std	min	max	q05	q25	q75	q95	skew	kurtosis	cv	rolling_vol_7	rolling_vol_30
EU	Інше/невідомо	other	linear	0	1478	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
EU	Інше/невідомо	other	pchip	0	1478	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
EU	Інше/невідомо	other	real	1056	422	nan	111.92024159890684	111.54602963226641	9.769089377787243	85.09705606113158	132.37006316419874	96.91861906080446	104.44988048442296	121.03481139576593	126.65109927513636	-0.0968182750340596	-0.7948157510925418	0.08728617127898212	0.0076768868159729576	0.008675645326450805
EU	Вершки	cream	linear	0	1478	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
EU	Вершки	cream	pchip	0	1478	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
EU	Вершки	cream	real	1056	422	nan	119.0249904616937	117.41427340175431	22.779287746811825	81.89986716652001	181.66680403065857	89.10613241317233	97.27939172416498	139.17423789310783	156.18510944158942	0.37639482185933626	-1.0528886551593721	0.19138239506217794	0.008983336386969399	0.010702317207885167
EU	Масло вершкове	butter	linear	0	1478	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
EU	Масло вершкове	butter	pchip	0	1478	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
EU	Масло вершкове	butter	real	1056	422	nan	262.92376012472243	251.9226003141099	59.42768694669505	175.25845388385454	373.9823955045714	189.26765577864094	207.2703788525187	325.4463978672138	355.8058803162048	0.3397037665829255	-1.3245924507639903	0.22602630860940257	0.007702665974553807	0.008515697455838888
EU	Молоко питне	milk	linear	0	1478	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
EU	Молоко питне	milk	pchip	0	1478	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
EU	Молоко питне	milk	real	1056	422	nan	28.473367809692313	29.618358542767858	5.142849814691726	15.829236887914286	35.88047252341785	17.758221086758162	26.081252316825	31.54852461989509	34.99451088020848	-0.9069650257955184	0.01306075486185021	0.18061965304087083	0.007344243801817971	0.008283589349468061
EU	Сир твердий	hard_cheese	linear	0	1478	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
EU	Сир твердий	hard_cheese	pchip	0	1478	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
EU	Сир твердий	hard_cheese	real	1056	422	nan	219.88132509752987	218.45870645677925	32.51664390005512	140.46569360260713	274.4434278616792	154.65888498191953	206.03342448541127	244.42855900697273	266.95436872353736	-0.5703651321774948	-0.2076985483003213	0.14788269938628096	0.007809487903776001	0.008715988438032821

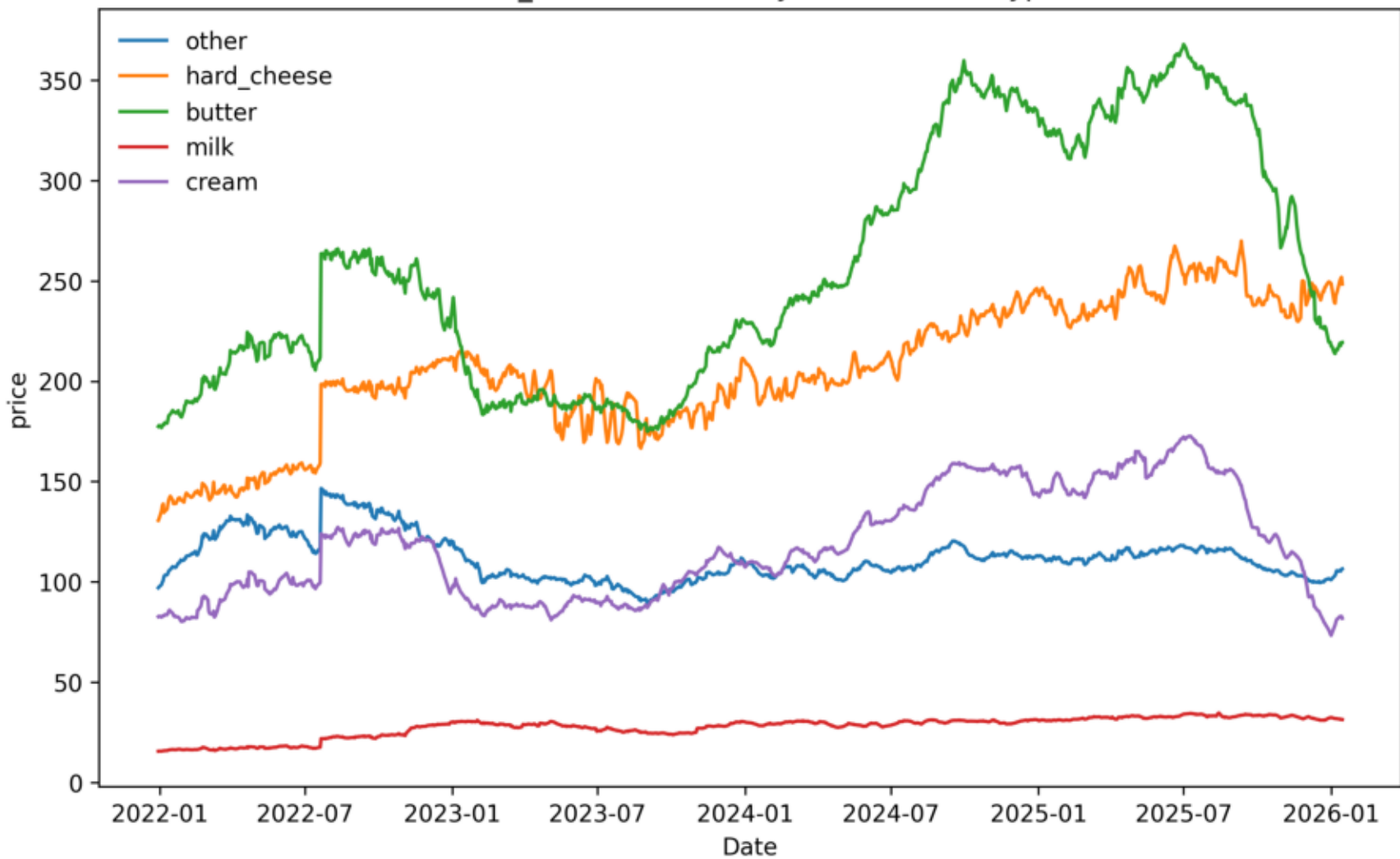
Table: tests

source	product	standardized_type	series_variant	n_obs	adf_p	kpss_p	ljungbox_p	bp_p	white_p	jb_p	stability_flag	stability_drift	action_label	integration_class	ac_risk	het_risk	non_normal_risk	stability_risk_class	recommended_action	recommended_model_family	avoid	diag_note	
EU	Інше/невідомо	other	linear	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Інше/невідомо	other	pchip	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Інше/невідомо	other	real	212	0.07737374247164855	0.038275403709735326	0.004697004424801457	0.4464181546898449	0.469485874904799	0.0	1	8.87118142148681	non-stationary: differenc...	I(1)-like	high	low	high	high	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Вершки	cream	linear	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Вершки	cream	pchip	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Вершки	cream	real	212	0.5600203781682339	0.01	0.34395405551648484	0.017579559638661577	5.977080082142508e-05	1.6313943834030816e-120	1	33.272649543316035	non-stationary: differenc...	I(1)-like	low	high	high	high	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Масло вершкове	butter	linear	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Масло вершкове	butter	pchip	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Масло вершкове	butter	real	212	0.5267216667834975	0.01	1.6293867941311686e-08	0.8578364022171215	0.32437351213264226	0.0	1	96.13800980318905	non-stationary: differenc...	I(1)-like	high	low	high	high	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Молоко питне	milk	linear	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Молоко питне	milk	pchip	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Молоко питне	milk	real	212	0.219904832363445	0.01	0.0289261007750083	0.47994041462283066	0.7065882516933646	0.0	1	7.260391990850234	non-stationary: differenc...	I(1)-like	high	low	high	high	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Сир твердий	hard_cheese	linear	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Сир твердий	hard_cheese	pchip	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...
EU	Сир твердий	hard_cheese	real	212	0.2161728016525135	0.01	0.5648061329308757	0.8550093177818229	0.4357472453890435	0.0	1	46.86892425275295	non-stationary: differenc...	I(1)-like	low	low	high	high	Use differences or cointe...	Diagnostics-dependent	Unvalidated	level OLS und...	Follow diagnostics-driven...

sheet_eu: Time Series by Product



sheet_eu: Time Series by Standardized Type



sheet_eu: Distribution of price

